

std::gcd | C++ inbuilt function for finding GCD

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In many competitive programming problems, we need to find greatest common divisor also known as **gcd**. Euclids algorithm to find gcd has been discussed [here](#). C++ has the built-in function for calculating GCD. This function is present in header file.

Syntax for C++14 :

Library: 'algorithm'

__gcd(m, n)

Parameter : m, n

Return Value : 0 if both m and n are zero,
else gcd of m and n.

Syntax for C++17 :

Library: 'numeric'

gcd(m, n)

Parameter : m, n

Return Value : 0 if both m and n are zero,
else gcd of m and n.

```
// CPP program to illustrate
// gcd function of C++ STL
#include <iostream>
#include <algorithm>
// #include<numeric> for C++17

using namespace std;

int main()
{
    cout << "gcd(6, 20) = " << __gcd(6, 20) << endl; // gcd(6,20)
    for C++17
}
```



Output

```
gcd(6, 20) = 2
```

Time Complexity: $O(\log n)$

Auxiliary Space: $O(1)$

Program to Find the gcd of all numbers in a vector.

```
#include<bits/stdc++.h>
#include <iostream>
#include <algorithm>
using namespace std;
int main()
{
    vector<int> numbers = { 12, 15, 18, 21, 24 };
    int ans = __gcd(numbers[0], numbers[1]);
    for (int i = 2; i < numbers.size(); i++)
    {
        ans = __gcd(ans, numbers[i]);
    }
    cout << "The GCD of the numbers in the vector is: "
    <<ans<<endl;
    return 0;
}
```

Output

```
The GCD of the numbers in the vector is: 3
```

Time Complexity: $O(k \cdot \log n)$

Auxiliary Space: $O(k)$

Note: If either M or N is not an integer type, or if either is (possibly cv-qualified) bool, the program is ill-formed. Also, If either |m| or |n| is not representable as a value of type `std::common_type_t`, the behavior is undefined.

```
// CPP program to illustrate
```

```
// undefined behavior of
// gcd function of C++ STL
#include <iostream>
#include <algorithm>
// #include<numeric> for C++17

using namespace std;

int main()
{
    cout << "gcd(2.0, 8) = " << __gcd(2.0, 8) << endl; //
gcd(2.0,8) for C++17
}
```

Output:

Error, As the data type float is not supported by std::gcd.

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